

csv2xyd 2.1 User Manual

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Introduction

Biodiversity repositories are essential for scientific research, providing valuable information for ecology, biogeography, and conservation studies. A key component of biogeographical research is identifying areas of endemism, which involves analyzing many datasets that frequently contain typographical errors, insufficient data, and taxonomic discrepancies; [csv2xyd](#) (Liria & Soto-Vivas, 2025) is a Python-based software designed to process large CSV files and convert them into XYD format, ideal for endemism analysis with [NDM/vNDM](#). It provides advanced functionalities for preprocessing, filtering, and combining biodiversity data while offering spatial analysis capabilities. This update now included the option for analyze output files produced by NDM/vNDM software called “[NDMOutAnalyzer](#)”.

System Requirements

- Python 3.12.4 or higher
- Required libraries: Tkinter 0.1.0, Pandas 2.2.2, Dask 2024.7.0, FuzzyWuzzy 0.18.0, Folium 0.17.0, NumPy 2.0.0, GeoPandas 1.0.1, Shapely 2.0.5, xlswriter 3.8, matplotlib 3.11.0
- Recommended hardware: 16 GB of RAM or more for handling large datasets.

Windows 10 / Linux (Ubuntu 22.04-2 LTS)

Installation

1. Install Python from python.org.
2. Install the necessary libraries by running command prompt:

```
pip install tkinter pandas dask fuzzywuzzy geopandas shapely folium numpy xlswriter matplotlib
```

3. Download the software from the GitHub repository and run the `csv2xyd` application.

Main Menu Overview

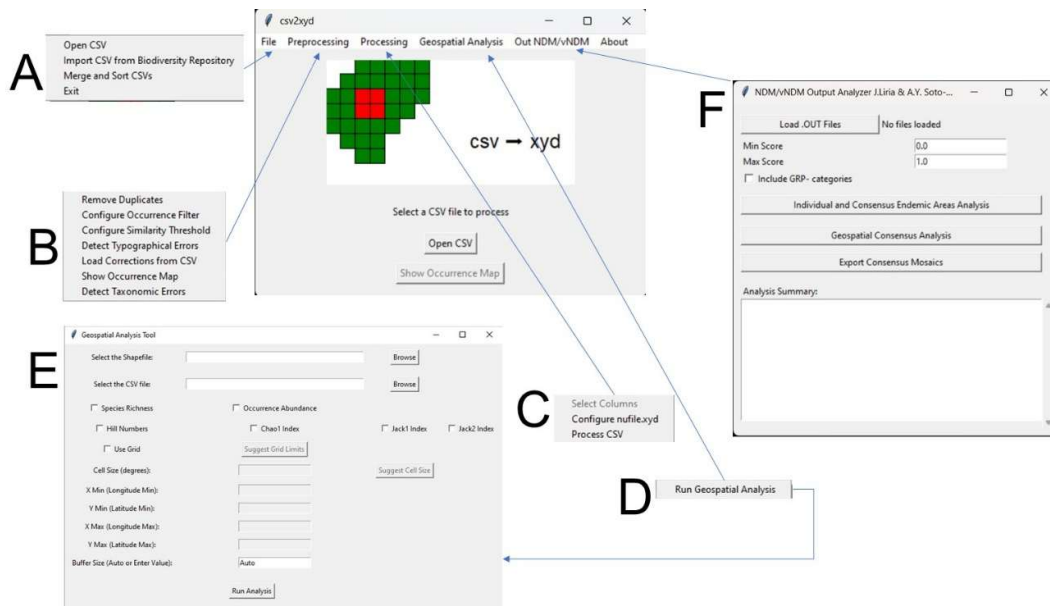


Figure 1. Main menu overview of csv2xyd: A) File, B) Preprocessing, C) Processing, D) Geospatial Analysis, and E) Details of the options in Geospatial Analysis, and F) Out NDM/vNDM post-analysis tools for output files, including consensus, endemism summaries, Beta diversity, and mosaics figure.

File Menu (Fig. 1A)

1. Open Semi-Formatted CSV

- Use this option to open a CSV file with essential columns like species (genus and species), latitude, and longitude. Additional columns for higher taxonomic levels (e.g., phylum, class, order) are also supported.
- Example: Open the file "World_Arthropoda.csv", choose the delimiter (comma or tab), and the program will display the number of occurrences and estimated memory usage.

2. Open CSV from a Biodiversity Repository

- Load CSV files from repositories like GBIF. The program allows you to choose the relevant columns (e.g., phylum, class, order, species) and apply filters such as valid coordinates or specific taxonomic groups.
- Example: Open "0024888-240626123714530.csv", select columns, filter for "Amphibia", and save the filtered file as "Chordata_amphibia.csv".
- This is a list of some commands for filters:

Command	Description	Use
texto	Includes rows/occurrences that contain exactly the specified text.	Insecta in the class column will include all rows where class field contains "Insecta" Insecta,Arachnida in the class column will include all rows where class field contains "Insecta" or "Arachnida".
!texto	Excludes rows/occurrences that contain exactly the specified text.	!cf. in the species column will exclude all rows where species field contains "cf." !.*cf.* in the species column will exclude all rows where species contains the "cf." text in any position.
textonly	Filters rows/occurrences where the column contains only text.	textonly in the phylum column will select rows where this taxonomic category contains text.
isnumeric	Filters rows/occurrences where the column contains only numeric values.	isnumeric in the longitude column will select rows where longitude contains numeric values.
.	Filters rows that contain any character.	.+ in a column will select all rows where the column contains any text.

3. Merge and Sort CSV Files

- This feature allows merging multiple CSV files and sorting the combined dataset by specific fields.
- Example: Merge "File_1.csv", "File_2.csv", etc., and sort them by genus and species, saving the result as, for example "File_merge.csv".

4. Reorder Columns

- After merging CSVs, you can rearrange the column order to suit your needs.

Preprocessing Menu (Fig. 1B)

1. Remove Duplicates

- Removes duplicate species entries with the same coordinates.

2. Occurrence Filter

- Set a minimum number of occurrences per species (default is 10), ensuring that only species with sufficient data are processed.

3. Detect Typographical Errors

- This option detects species names with small differences, which may indicate typos. A similarity threshold (default: 90%) can be adjusted. The program generates a report (possible_typo_errors.csv) for review and correction.

4. Show Occurrence Map

- Displays a random selection of occurrences on an interactive map using Folium. The default sample size is 10%, but this can be adjusted.

5. Detect Taxonomic Errors

- Identifies species with inconsistent taxonomic assignments, generating a report for review.

Processing Menu (Fig. 1C)

1. Process CSV for XYD

- Convert a CSV file into the XYD format required for NDM/vNDM analysis. Select columns (e.g., genus and species), define grid limits (gridx, gridy), and fill values to generate the XYD file.

2. Export to TNT

- The program also generates a [TNT](#) file that can be processed with `gettaxo.run` macro to include higher groups in the endemism analysis.

Geospatial Analysis Menu (Figs. 1D,E)

1. Run Geospatial Analysis

- Perform exploratory spatial analysis on species occurrences and polygons. Load a shapefile or generate a custom grid to match the XYD file's boundaries.
- Adjust the grid size and analyze richness, diversity, and species evenness across the spatial distribution of data.

Out NDM/vNDM Menu (Fig. 1F) *New menu*

This menu provides tools for post-processing and analyzing output files generated by NDM/vNDM, allowing users to summarize endemic species, individual areas, consensus areas, and their spatial relationships across different grid resolutions.

1. Analyze NDM/vNDM Output (.out)

This option opens the NDM/vNDM Output Analyzer module, a dedicated interface for processing one or multiple `.out` files generated by NDM or vNDM.

Main functionalities include:

- Loading multiple `.out` files corresponding to different grid sizes or analytical runs.
- Filtering endemic species by minimum and maximum endemism score.
- Optional inclusion or exclusion of supraspecific categories (GRP-prefix).
- Automatic extraction and summary of:
 - Individual endemic areas (SET NR.)
 - Consensus areas
 - Species represented in each analysis
 - Endemic species per area and per consensus
- Export of all results to a multi-sheet Excel file, facilitating downstream analyses and reporting.

2. Geospatial Consensus Analysis

This option performs a spatial analysis of consensus areas, based on polygon (WKT files) and point files previously exported from NDM/vNDM analyses. The analysis requires folders containing paired files in CSV format:

- **conX_N.csv** → polygon representation of consensus areas
- **pointX_N.csv** → species occurrence points associated with each consensus

Each pair corresponds to a consensus area X derived from N individual areas.

Main outputs include:

- **Consensus summary:** Area size, number of species, polygon area, and grid resolution.
- **Consensus overlap analysis:** Pairwise spatial overlap between consensus areas, including intersection area and Jaccard index.
- **Shared species and genera:** Identification of shared and exclusive species and genera among consensus areas across different grid sizes.
- **Beta Diversity Partitioning** (Baselga, 2009): Advanced decomposition of biotic dissimilarity between areas, calculating:
 - **Turnover** (β_{sim}): Species replacement between areas.
 - **Nestedness** (β_{sne}): Dissimilarity resulting from species loss/gain (subsets).
 - **Total Dissimilarity** (β_{sor} and β_{jac}): Overall taxonomic difference.

All results are exported to a single Excel file with multiple sheets (including specialized sheets for genus-level summaries and analytical statistics), enabling comparative spatial analyses and integration with GIS software.

3. Export Consensus Mosaics

This module provides a interface for creating high-quality, multi-panel figures for publication. It allows for the visual comparison of consensus areas across different resolutions or treatments.

Main functionalities include:

- **GIS Base map integration:** Allows the loading of external vector layers (Shapefile .shp or GeoPackage .gpkg) to provide geographic context such as political borders, ecoregions, or topography.
- **Automated grid visualization:** Generates a synchronized mosaic where each panel displays a specific consensus area. The system ensures spatial consistency by maintaining the same geographic extent and scale across all panels.
- **Publication-Ready export:** Exports the final composition (mosaic) as a high-resolution PNG file (300 DPI), suitable for scientific journals.

Example Workflow

1. **Opening and Filtering a CSV:** Load a CSV file, select relevant columns (e.g., species, latitude, longitude), and apply filters (e.g., valid coordinates or a specific taxonomic group). Open CSV file downloaded from biodiversity repositories (e.g., GBIF, CRIA, etc).
2. **Merging Datasets:** Combine multiple CSV files (e.g., merge form different sources datasets).
3. **Preprocessing:** Remove duplicates, select occurrence filter, detect typographical and taxonomic errors, and visualize a random subset of occurrences on a map.
4. **Exporting XYD:** Once preprocessing is complete, convert the data into XYD format for use in NDM/vNDM.
5. **Geospatial Analysis:** Run spatial analyses using custom grids or shapefiles, adjusting the grid size to optimize the spatial representation of your data.
6. **Analyze NDM/vNDM Output:** Analyze NDM/vNDM output files to summarize endemic species, individual and consensus areas, and evaluate spatial overlap among consensus areas, exporting all results to multi-sheet Excel files.

Contact and Support

For support or to contribute to the development of **csv2xyd-NDMOutAnalyzer**, visit the GitHub repository or contact the lead developer at jonathan.liria@gmail.com

Licensing

The software is open-source and licensed under MIT.

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